

STUDENT

SNO	NAME	AGE
S1	N1	21
S2	N2	22
S3	N3	23
S4	N4	24

COURSE

Sno	Cname	Fees
S1	C1	1000
S3	C3	3000
S5	C5	5000

Assume join condition: STUDENT.SNO = COURSE.Sno.

1. LEFT OUTER JOIN (STUDENT LEFT JOIN COURSE)

SQL idea:

```
SELECT *  
FROM STUDENT  
LEFT JOIN COURSE  
ON STUDENT.SNO = COURSE.Sno;
```

Rule:

- Take **all rows from STUDENT**.
- Match rows from COURSE where SNO is equal.
- If no match, fill COURSE columns with NULL.
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Step-by-step

Step 1 – Start with first row of STUDENT (S1):

- STUDENT row: (S1, N1, 21).
- Look in COURSE for Sno = S1 → found (S1, C1, 1000).
- Output row: (S1, N1, 21, C1, 1000).

Step 2 – Second row of STUDENT (S2):

- STUDENT row: (S2, N2, 22).
- Look in COURSE for Sno = S2 → **no row**.
- Output row with NULLs for COURSE: (S2, N2, 22, NULL, NULL).

Step 3 – Third row of STUDENT (S3):

- STUDENT row: (S3, N3, 23).
- COURSE Sno = S3 → found (S3, C3, 3000).
- Output: (S3, N3, 23, C3, 3000).

Step 4 – Fourth row of STUDENT (S4):

- STUDENT row: (S4, N4, 24).
- COURSE Sno = S4 → no match.
- Output: (S4, N4, 24, NULL, NULL).

Final LEFT JOIN result

SNO	NAME	AGE	Cname	Fees
S1	N1	21	C1	1000
S2	N2	22	NULL	NULL
S3	N3	23	C3	3000
S4	N4	24	NULL	NULL

Notice: all 4 students appear; S2 and S4 have no course data.

2. RIGHT OUTER JOIN (STUDENT RIGHT JOIN COURSE)

SQL idea:

```
SELECT *  
FROM STUDENT  
RIGHT JOIN COURSE  
ON STUDENT.SNO = COURSE.Sno;
```

Rule:

- Take **all rows from COURSE**.
- Match rows from STUDENT where SNO is equal.
- If no match, fill STUDENT columns with NULL.

Step-by-step

Step 1 – First row of COURSE (S1):

- COURSE row: (S1, C1, 1000).
- Find STUDENT SNO = S1 → (S1, N1, 21).
- Output: (S1, N1, 21, C1, 1000).

Step 2 – Second row of COURSE (S3):

- COURSE row: (S3, C3, 3000).
- Find STUDENT SNO = S3 → (S3, N3, 23).
- Output: (S3, N3, 23, C3, 3000).

Step 3 – Third row of COURSE (S5):

- COURSE row: (S5, C5, 5000).
- STUDENT SNO = S5 → **no row**.
- Output: (NULL, NULL, NULL, C5, 5000).

Final RIGHT JOIN result

SNO	NAME	AGE	Cname	Fees
S1	N1	21	C1	1000

S3	N3	23	C3	3000
NULL	NULL	NULL	C5	5000

Notice: all 3 courses appear; S5 has no matching student.

3. FULL OUTER JOIN

SQL idea (conceptual):

```
SELECT *
FROM STUDENT
FULL OUTER JOIN COURSE
ON STUDENT.SNO = COURSE.Sno;
```

Rule:

- Take **all rows from both tables**.
- Where SNO matches, merge into one row.
- Where a row has no match in the other table, show NULLs for missing side.

We can think of it as:

- Start with LEFT JOIN result.
- Add the COURSE rows that didn't match any STUDENT (i.e., S5).

Step-by-step

Start from LEFT JOIN rows (already computed):

1. (S1, N1, 21, C1, 1000) – match
2. (S2, N2, 22, NULL, NULL) – student only
3. (S3, N3, 23, C3, 3000) – match
4. (S4, N4, 24, NULL, NULL) – student only

Now check COURSE rows for any Sno not already matched:

- COURSE S1 and S3 are already matched.
- COURSE S5 is **unmatched**, so we add:

5. (NULL, NULL, NULL, C5, 5000) – course only

Final FULL OUTER JOIN result

SNO	NAME	AGE	Cname	Fees
S1	N1	21	C1	1000
S2	N2	22	NULL	NULL
S3	N3	23	C3	3000
S4	N4	24	NULL	NULL
NULL	NULL	NULL	C5	5000

- Rows with **S1, S3**: present in both tables → combined.
- Rows with **S2, S4**: only in STUDENT → COURSE columns NULL.
- Row with **S5**: only in COURSE → STUDENT columns NULL.